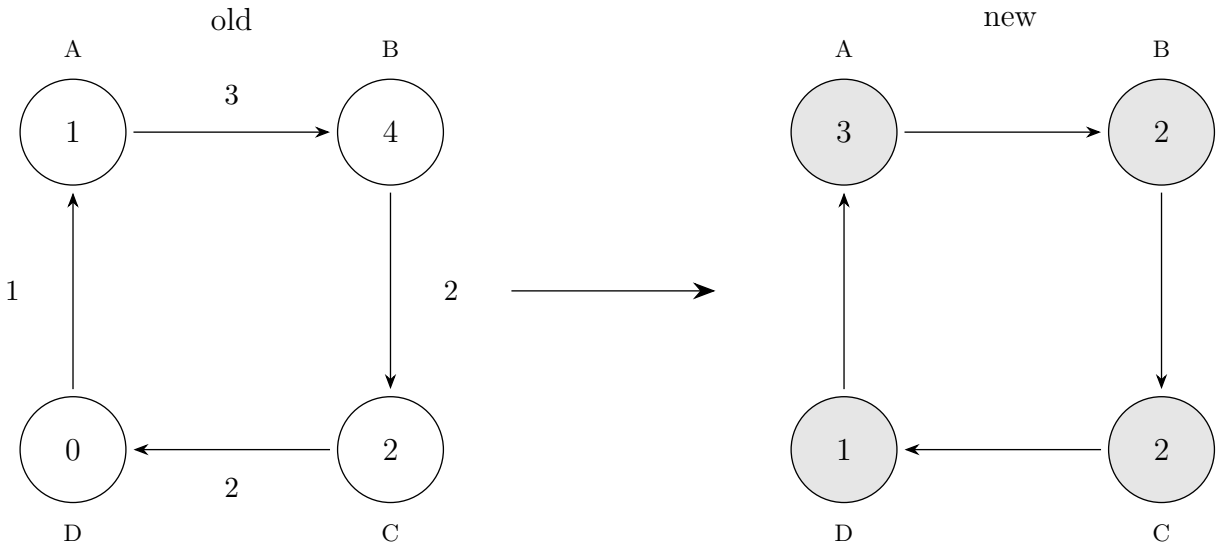
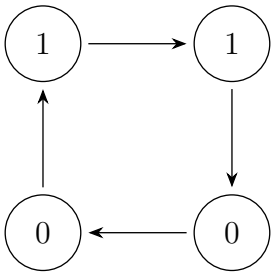
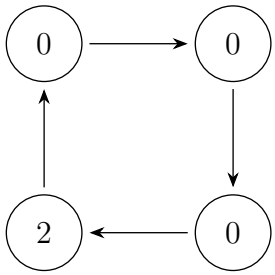
Use whole-number heights, including 0. Follow each arrow: the gap between its two old heights becomes the new height at its tail. Make every new tower before changing any old tower. An all-zero start takes 0 moves.

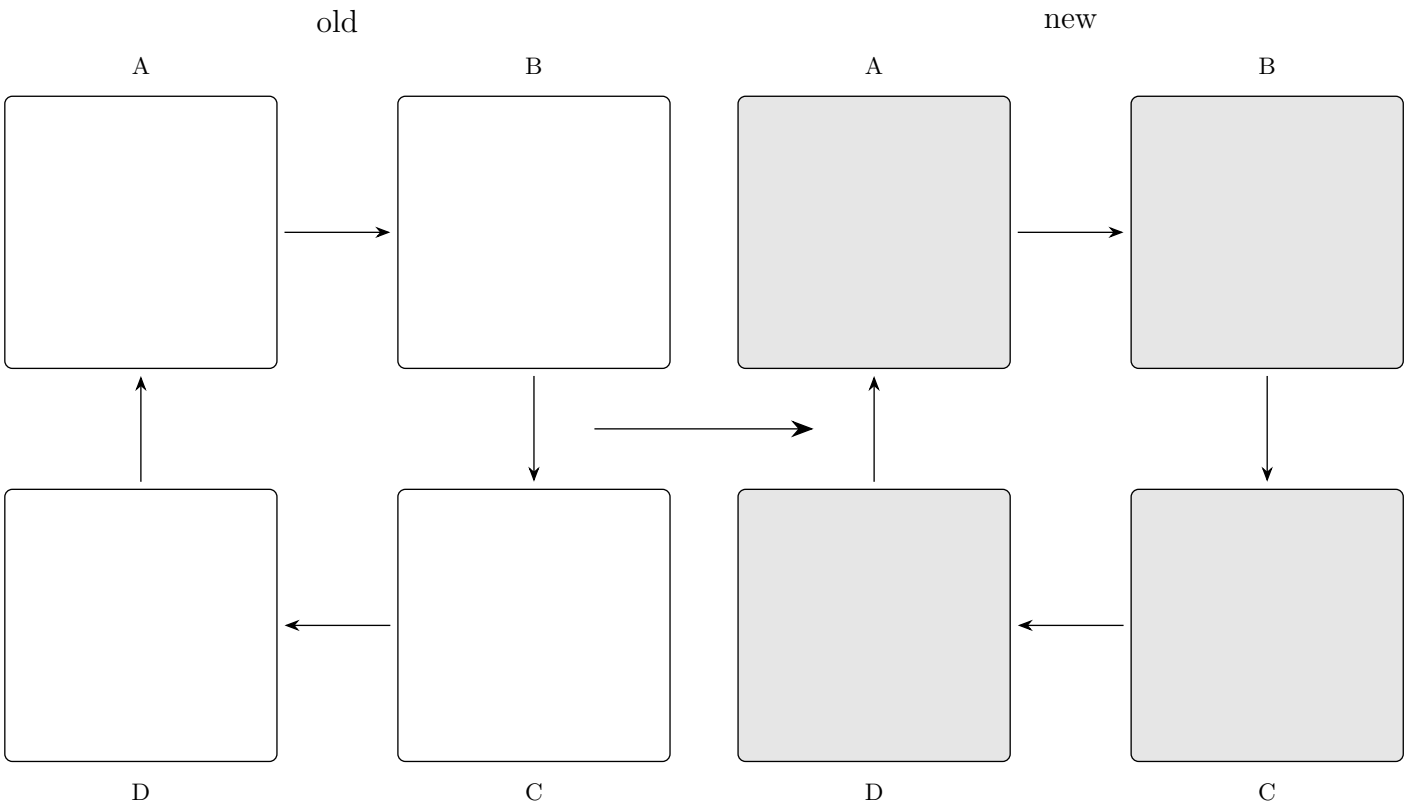


Edge A to B has gap 3, so new A is 3.

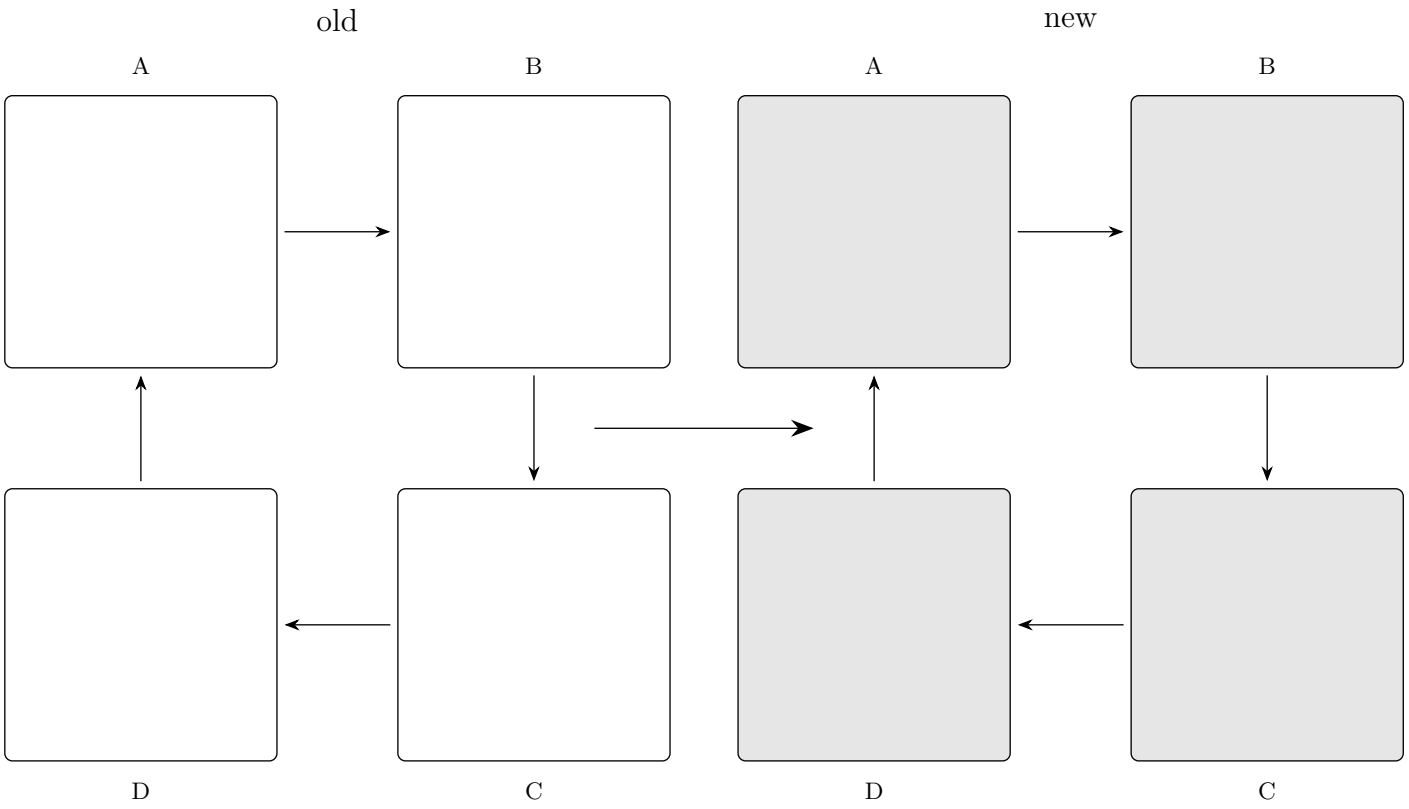
Problem 1: Run each start until all four heights are 0. Which lasts longer?

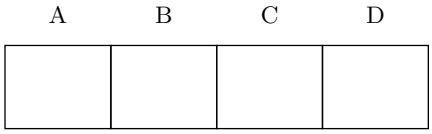




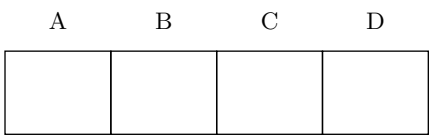


Problem 2: Choose starting heights from 0 to 5. Find a ring that lasts as many moves as you can. Count the move that first makes all four heights 0.



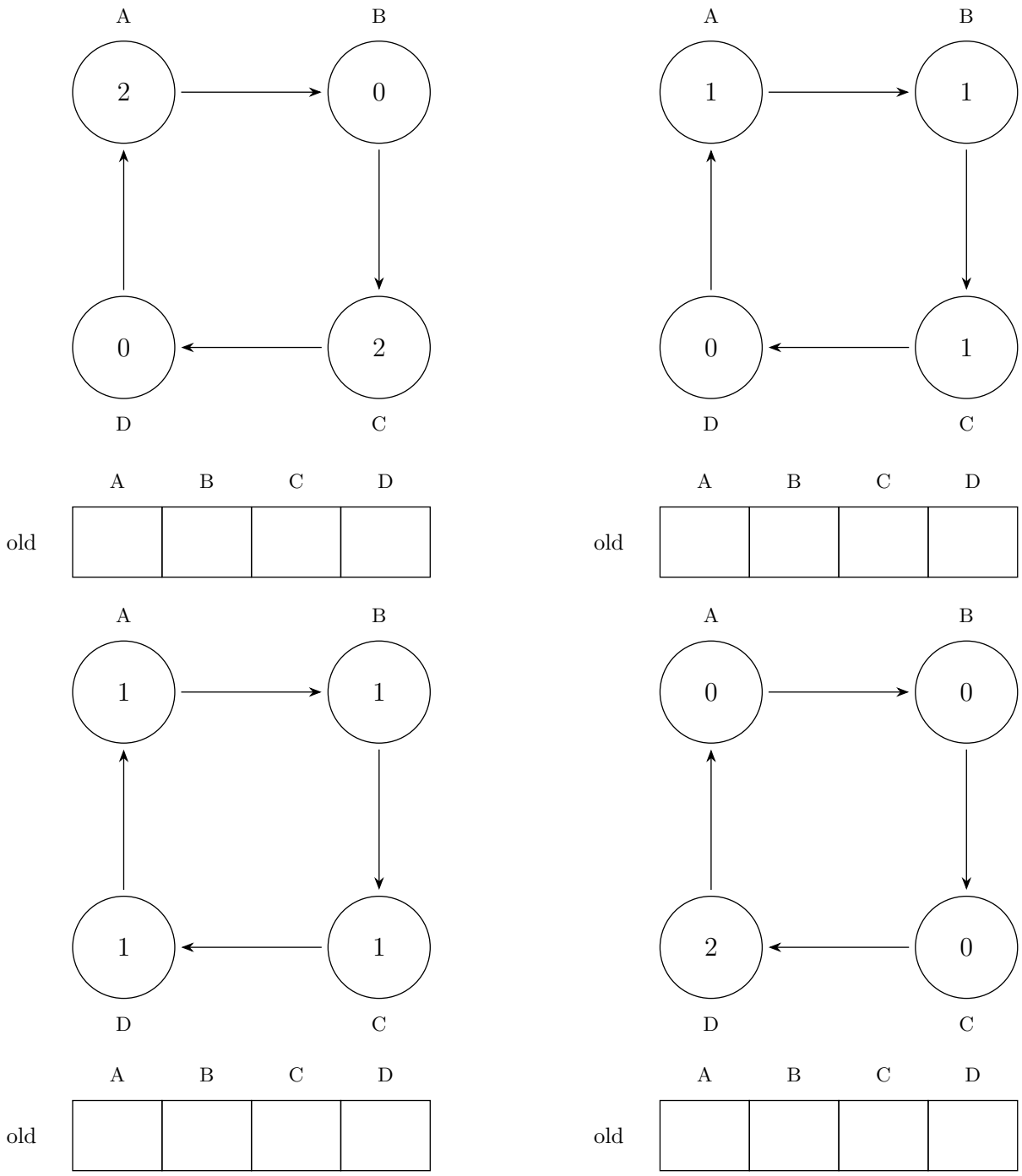


Moves to zero: _____

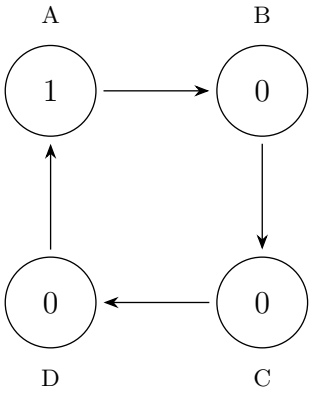


Moves to zero: _____

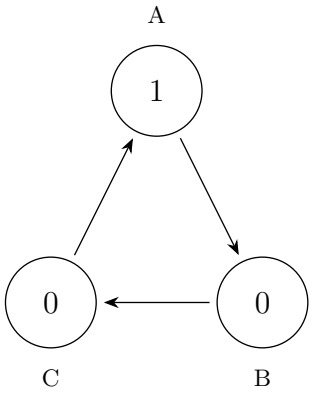
Problem 3: Could each ring be the result of one move? Find an old ring that makes it, or explain why no old ring can work.



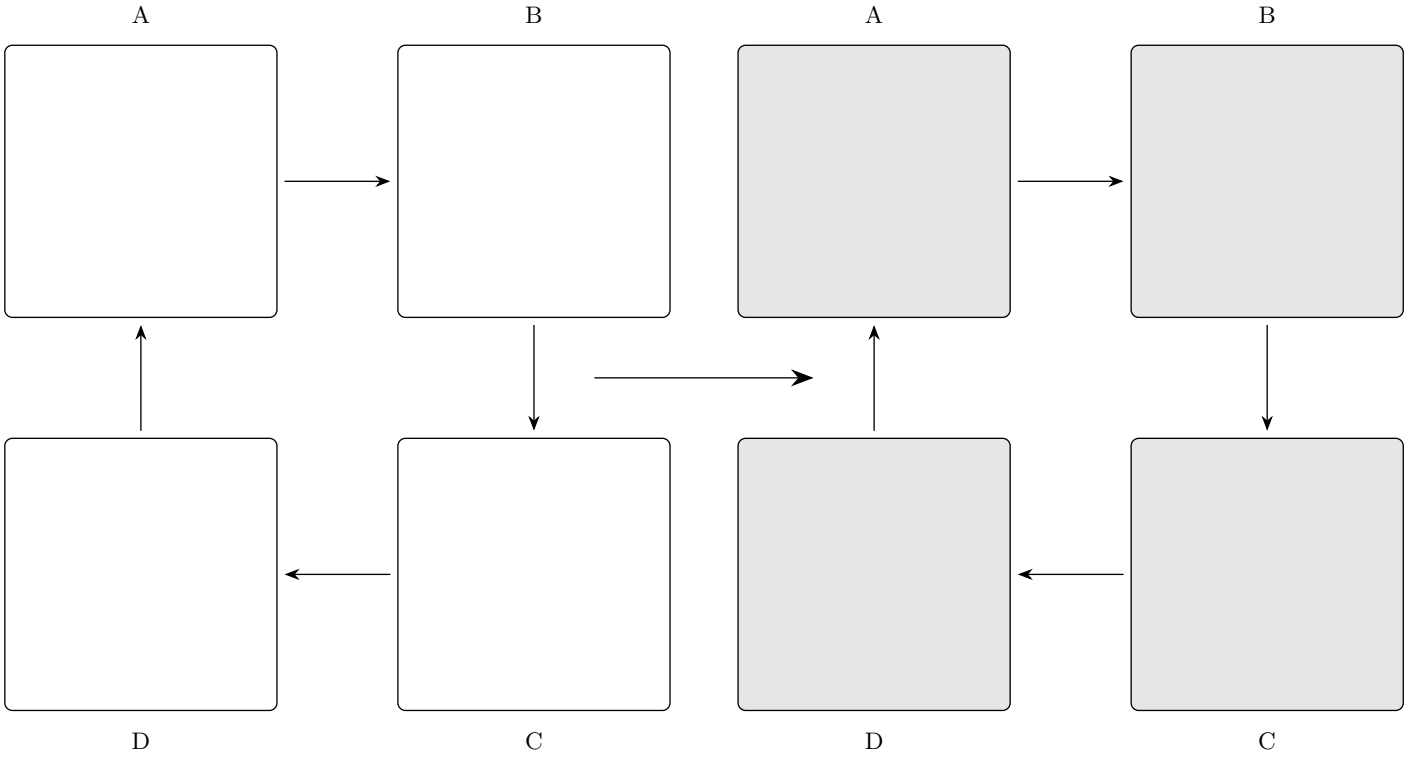
Problem 4: Use the gap rule on both rings. Keep going until the whole ring repeats or becomes all zeros. Does the number of stations change what can happen?



old

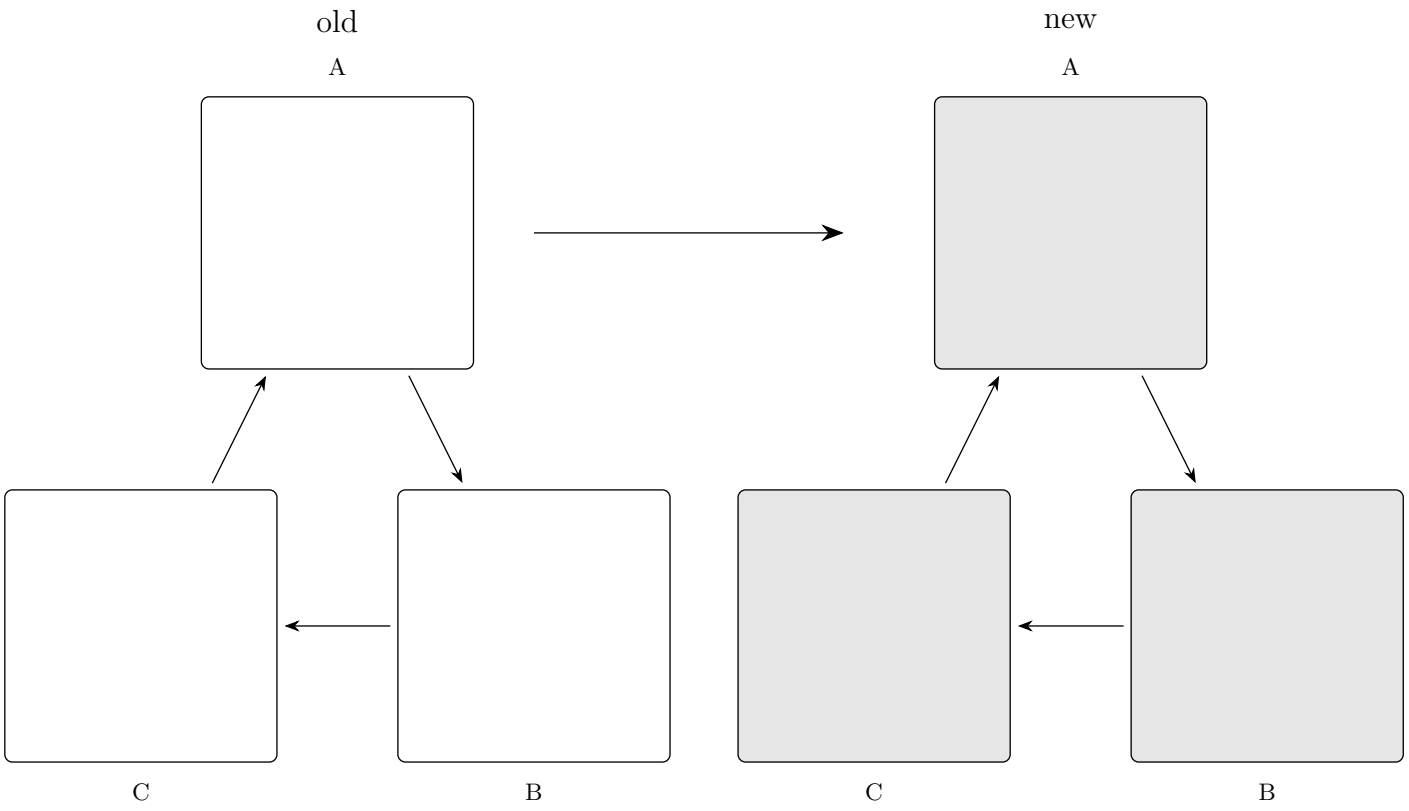


new



	A	B	C	D
1				
2				
3				

	A	B	C	D
4				
5				
6				



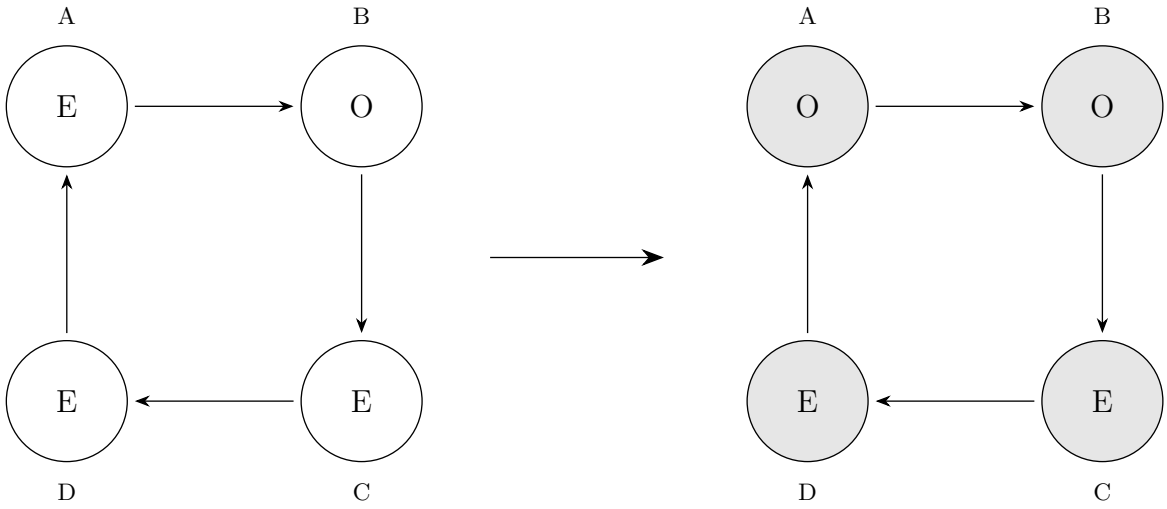
	A	B	C
1			
2			
3			

	A	B	C
4			
5			
6			

Problem 5: Replace even heights by E and odd heights by O. Use this two-letter gap rule. Which letter rings are possible after four moves? Explain why you have checked every kind of start.

E and E give E. O and O give E.

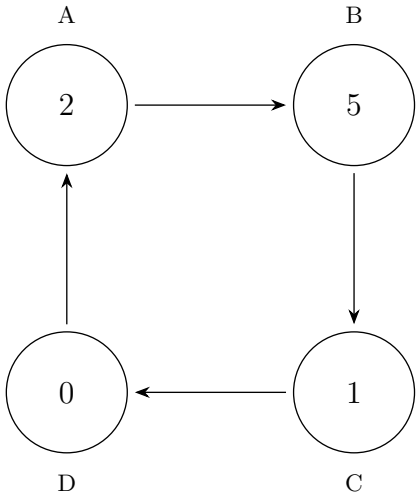
E and O give O. O and E give O.

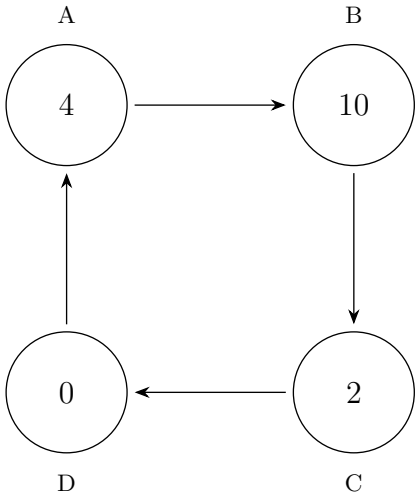


move	A	B	C	D
0				
1				
2				
3				
4				

A	B	C	D

Problem 6: Compare these two starts through several moves. What happens if all starting heights are doubled? Does your explanation also work for tripling?





Problem 7: Could any new height exceed the largest starting height? Explain why your answer holds however many moves are made.

Problem 8: Do four nonnegative whole-number heights always reach all zeros? Explain why every such start finishes, or find a start that never reaches zero.